SOFTWARE TESTING PROJECT

BYEM GYM Web-Based Management System

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Course: Software Testing
Project Type: Group Project

Github: https://github.com/burakkaygusuzoglu/BYEM-GYM-Software-Testing-Project

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1. Introduction

1.1 Project Overview

BYEM GYM is a modern web-based gym management system developed to digitalize common gym operations and improve user experience. The system allows users to create accounts, purchase memberships, reserve fitness classes, and manage their profiles through a user-friendly online platform.

The project was selected because it represents a realistic business application with multiple functional modules, role-based access control, booking logic, and user interaction scenarios. This makes it highly suitable for a comprehensive software testing project.

1.2 Purpose of the Testing Project

The purpose of this project is to design and execute a structured software testing strategy for the BYEM GYM system.

The testing process aims to evaluate:

- Functional correctness
- Input validation quality
- Booking and membership logic
- Access control mechanisms
- User experience reliability
- Defect detection and reporting
- Basic automation opportunities
- Basic performance behavior

The final objective is to ensure that the system meets acceptable quality standards before production use.

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2. System Description

BYEM GYM includes the following core modules:

Public Modules

- Home Page
- About Page
- Membership Plans
- Classes Page
- Contact Page

Authentication Modules

- User Registration
- User Login
- Logout

Member Features

- Membership Purchase
- Payment Simulation
- Class Booking
- Booking Cancellation

- Member Dashboard

Administrative Features

- Role-based access control
- Protected page restrictions

The system is designed for modern web browsers and supports responsive usage on desktop and mobile devices.

3. Test Plan

3.1 Objectives

The primary objectives of this testing project are:

- To verify that all major functionalities of the BYEM GYM system operate correctly
- To validate user input handling and form behavior
- To ensure membership and booking logic works as expected
- To verify access control between guest and authenticated users
- To identify and document system defects with proper severity levels
- To improve overall software quality and reliability

3.2 Scope of Testing

The following modules are included in the testing scope:

Authentication Module

- User registration
- User login
- Logout functionality

Membership Module

- Membership plan selection
- Payment simulation
- Membership activation status

Booking Module

- Class reservation
- Duplicate booking prevention
- Capacity control
- Booking cancellation

Dashboard Module

- Membership information display
- Booking summary
- User session behavior

Security / Access Control

- Guest restrictions
- Protected page access
- Logout session validation

Out of Scope

The following areas are not included:

- Real payment gateway integration
- Backend server penetration testing
- Third-party integrations
- Native mobile application testing

3.3 Testing Strategy

A structured multi-level testing approach will be applied.

Functional Testing

Used to verify whether features work according to requirements.

Validation Testing

Used to verify user input fields such as email, password, and required forms.

Negative Testing

Used to test invalid or unexpected user actions.

Boundary Value Testing

Used to test edge values such as minimum and maximum password length.

Role-Based Access Testing

Used to verify authorization restrictions.

Regression Testing

Used after defect fixes to ensure previous features still work correctly.

UI Testing

Used to verify visible interface behavior and responsiveness.

3.4 Test Environment

Component	Details
Operating System	Windows 10 / 11
Browser	Google Chrome
Frontend	HTML / CSS / JavaScript
Storage	LocalStorage
IDE	Visual Studio Code
Testing Type	Manual + Automated Testing

3.5 Tools Used

- Browser Developer Tools
- Manual Testing Checklists
- Excel / Word Documentation
- Selenium
- Apache JMeter

3.6 Entry Criteria

Testing begins when:

- The BYEM GYM system is fully implemented
- Main pages are accessible
- Core features are functional
- Test environment is ready

3.7 Exit Criteria

Testing is considered complete when:

- All planned test cases are executed
- Critical defects are resolved or documented
- Major modules operate successfully
- Final report is completed

3.8 Risks and Assumptions

Risks

- LocalStorage limitations
- Browser compatibility issues
- Missing edge-case validations

Assumptions

- Users have internet browser access
- Payment system is simulated only
- System is used in standard browser conditions

Test Strategy Summary

The table below summarizes all testing types applied across the BYEM GYM project, mapping each strategy to its objective and the test cases that exercised it.

#	Test Type	Objective	Test Cases	Result
1	**Functional Testing**	Verify core features work per requirements (registration, login, payment, booking, cancellation, logout).	TC-01, TC-03, TC-05, TC-06, TC-08, TC-09	**Passed**
2	**Negative Testing**	Confirm system rejects invalid inputs and duplicate actions.	TC-02, TC-04, TC-07	**Passed**
3	**Validation Testing**	Validate input fields (email format, password rules, required fields).	TC-01, TC-02, TC-03	**Passed**
4	**Boundary Value Testing**	Test edge inputs (min/max password length).	Applied within TC-01–TC-04	**Passed**
5	**Role-Based Access Testing**	Verify guest vs. authenticated user restrictions and protected pages.	TC-08, ST-01	**Mixed**
6	**UI Testing**	Inspect interface behavior, layout, and responsiveness.	BR-01, BR-02, BR-03, BR-04	**Issues found**
7	**Regression Testing**	Re-run prior tests after fixes to confirm no breakage.	Applied after defect fixes	**Passed**
8	**Automation Testing**	Automate repetitive auth flows with Selenium WebDriver.	AT-01, AT-02, AT-03	**Passed**
9	**Performance Testing**	Measure response time, throughput, and stability under load (Apache JMeter).	PT-01, PT-02, PT-03, PT-04	**3/4 Passed**
10	**Security Testing**	Validate session termination, access control, and data integrity risks.	ST-01, BR-05, BR-06, BR-07, BR-08	**Risks identified**

TC-01

Test Case ID: TC-01
Module: Registration
Title: Valid User Registration
Type: Functional

Preconditions: Register page open

Steps:

1. Enter valid data
2. Click Register

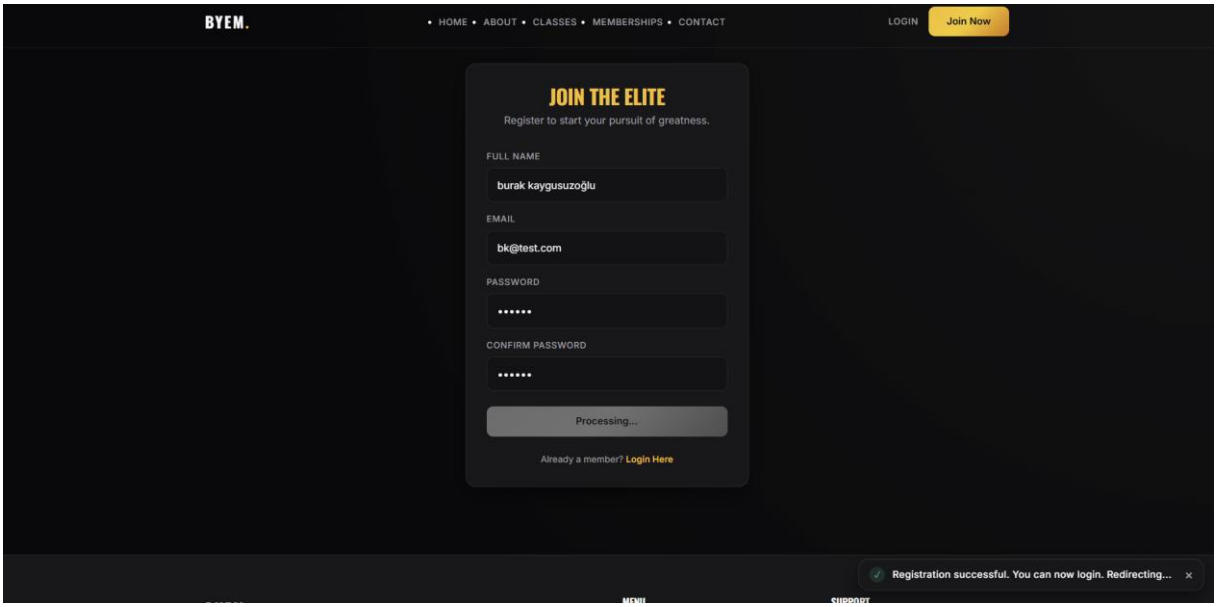
Test Data: bk@test.com / password:123456

Expected Result: User account created successfully.

Actual Result: User account was created successfully. You can now login.

Status: PASS

Screenshot:



TC-02

Test Case ID: TC-02
Module: Registration
Title: Existing Email Registration
Type: Negative Testing

Preconditions:

- Registration page open
- Test email already exists in system

Steps:

1. Open Register page
2. Enter valid full name
3. Enter an already registered email
4. Enter valid password
5. Confirm password
6. Click Register

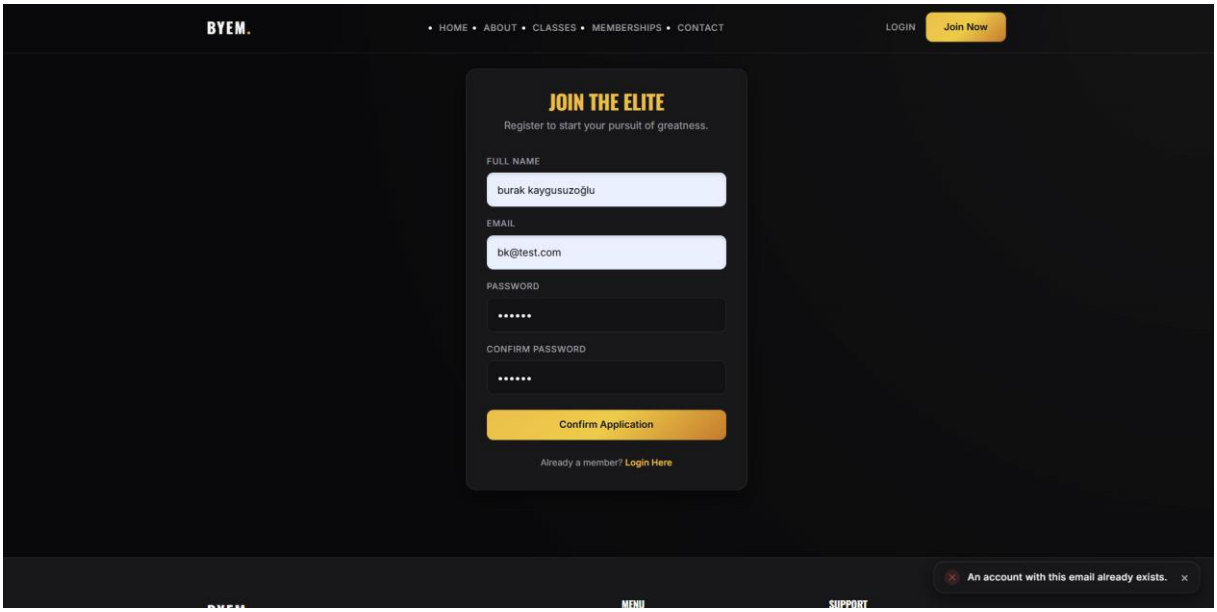
Test Data:
bk@test.com / password:123456

Expected Result:
System should reject registration and display duplicate email error message.

Actual Result:
An account with this email already exists.

Status: PASS

Screenshot:



TC-03

Test Case ID: TC-03
Module: Login
Title: Valid Login
Type: Functional Testing

Preconditions:

- Registered user exists
- Login page open

Steps:

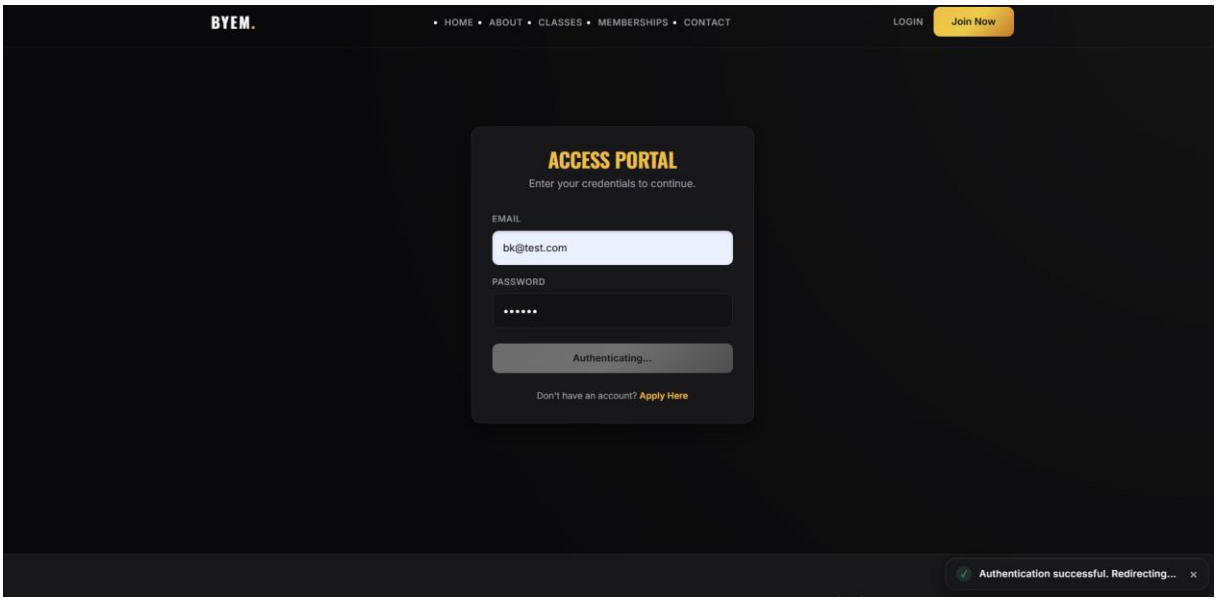
1. Open Login page
2. Enter valid email
3. Enter valid password
4. Click Login

Test Data:
bk@test.com / 123456

Expected Result:
User should log in successfully and be redirected to dashboard page.

Actual Result:
AUTH SUCCESSFUL. REDIRECTING --

Status: PASS
Screenshot:



TC-04

Test Case ID: TC-04
Module: Login
Title: Wrong Password Login
Type: Negative Testing

Preconditions:

- Registered user exists
- Login page open

Steps:

1. Open Login page
2. Enter valid email
3. Enter incorrect password
4. Click Login

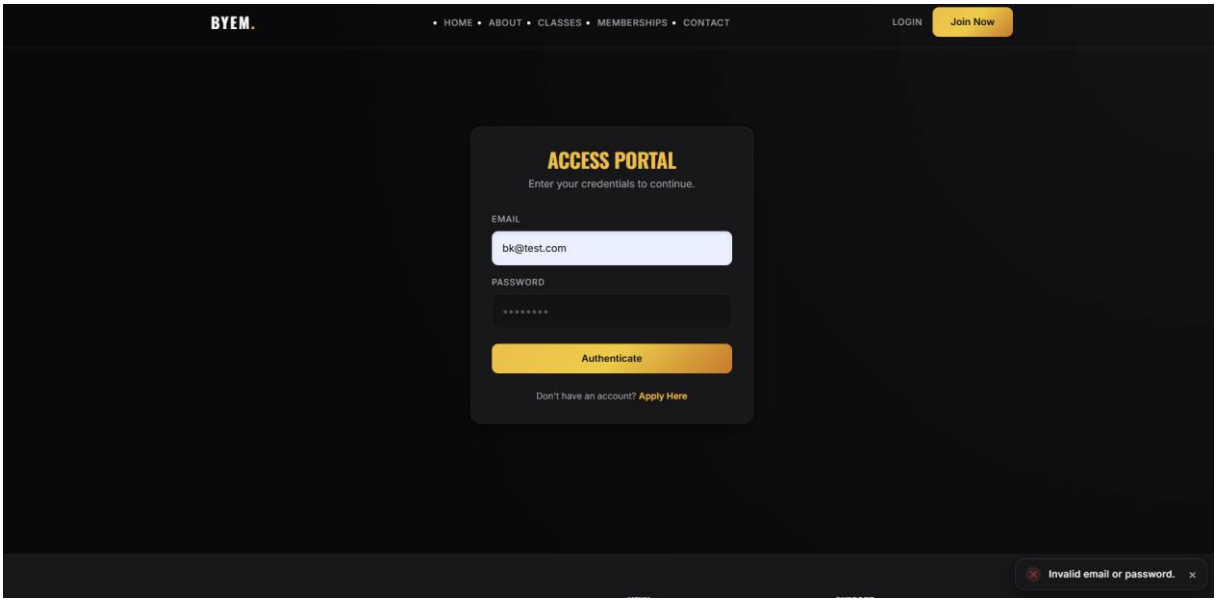
Test Data:
bk@test.com / wrong123

Expected Result:
System should deny login and display an error message.

Actual Result:
Invalid email or password

Status: PASS

Screenshot:



TC-05

Test Case ID: TC-05
Module: Payment
Title: Successful Payment
Type: Functional Testing

Preconditions:

- Membership plan selected
- Payment page open

Steps:

1. Enter valid payment information
2. Click Confirm Payment

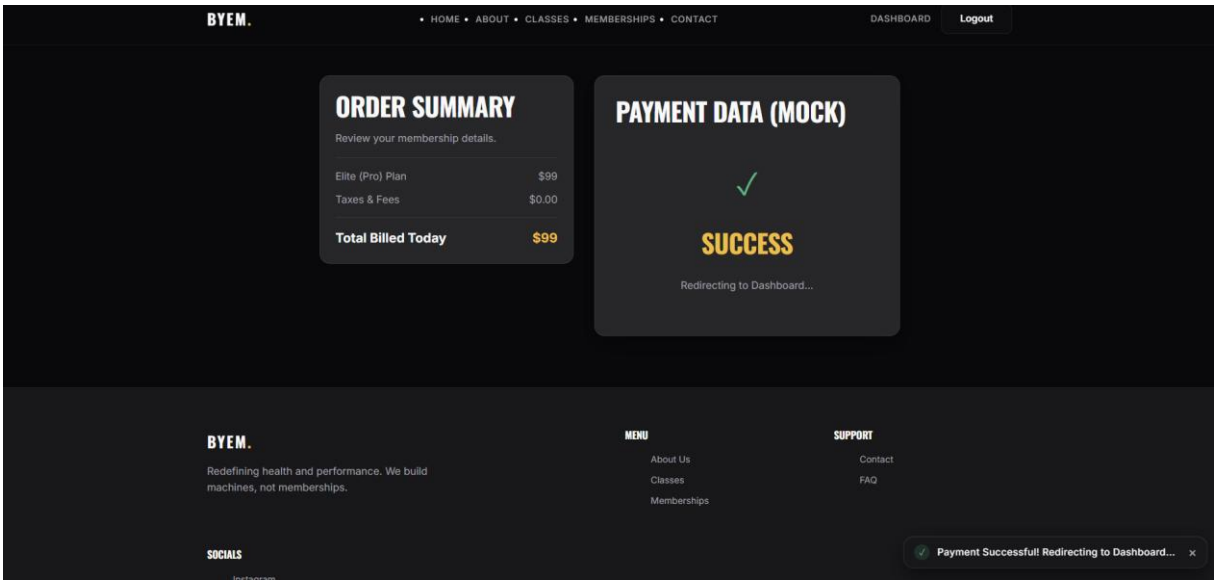
Test Data:
Valid card information

Expected Result:
Payment completed successfully and membership becomes active.

Actual Result: Payment successful – Redirecting to Dashboard___

Status: PASS

Screenshot:



TC-06

Test Case ID: TC-06
Module: Booking
Title: Booking with Active Membership
Type: Functional Testing

Preconditions:

- User is logged in
- Active membership exists
- Classes page open

Steps:

1. Open Classes page
2. Select an available class
3. Click Book

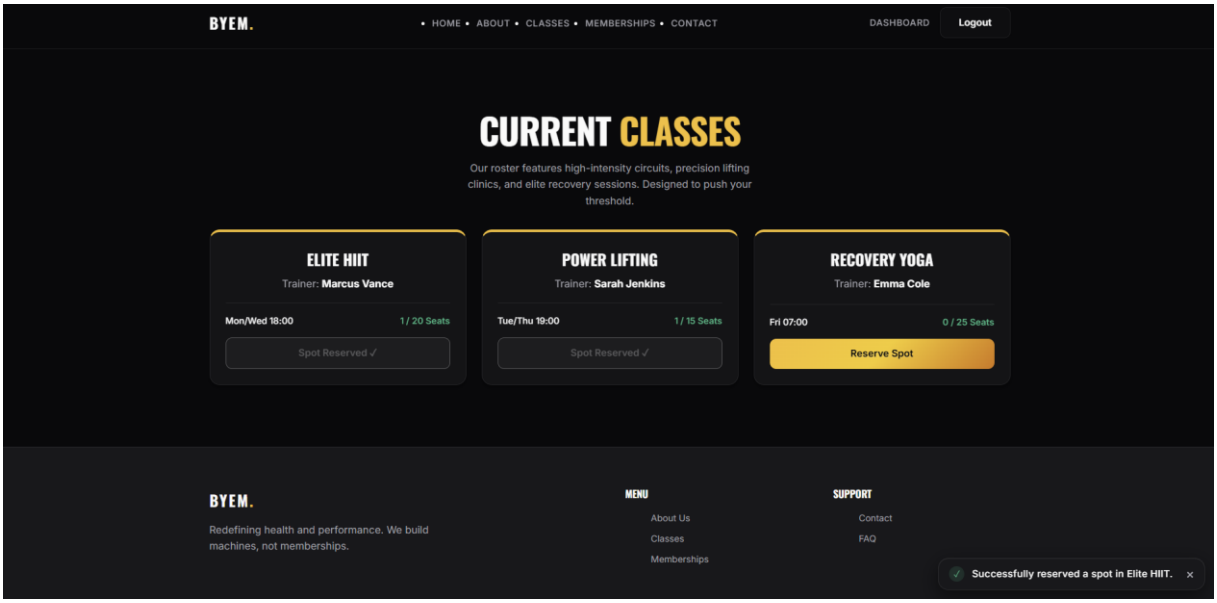
Test Data:
Any available class

Expected Result:
Class booking should be completed successfully.

Actual Result:
Successfully reserved a spot in _____.

Status: PASS

Screenshot:



TC-07

Test Case ID: TC-07
Module: Booking
Title: Duplicate Booking Attempt
Type: Negative Testing

Preconditions:

- User is logged in
- Same class already booked

Steps:

1. Try to book the same class again

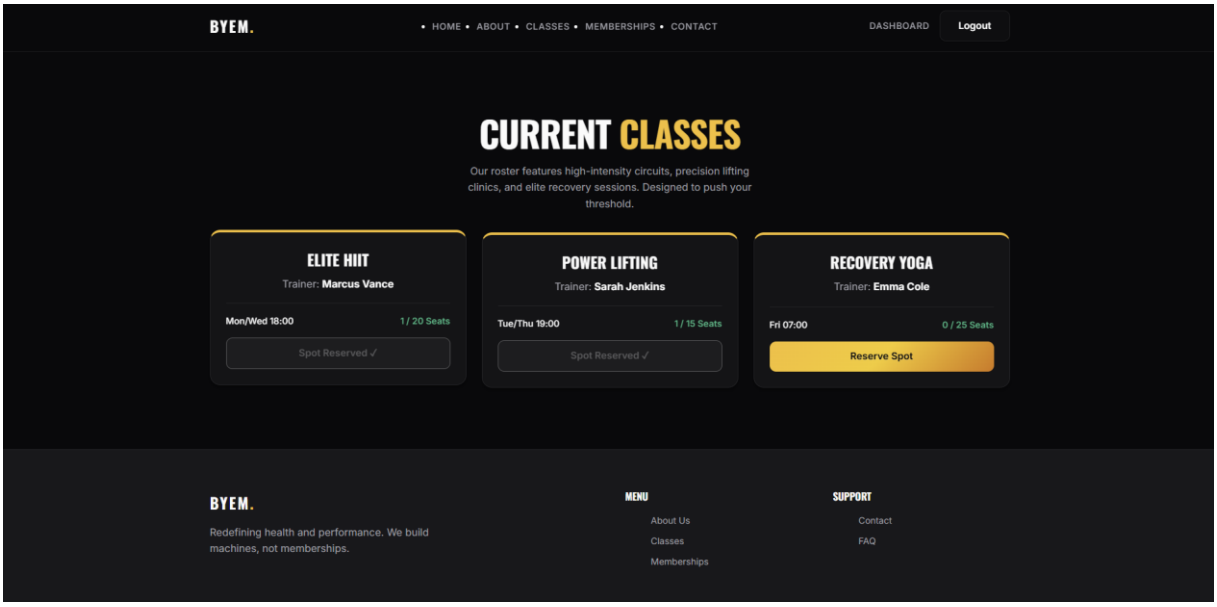
Test Data:
Previously booked class

Expected Result:
System should reject duplicate booking.

Actual Result:
When you reserved the spot, you cannot reserve the same spot again.

Status: PASS

Screenshot:



TC-08 / SECURİTY TESTİNG FOR ACCESS DASHBOARD AFTER LOG OUT!

Test Case ID: TC-08
Module: Session
Title: Logout Test
Type: Functional Testing

Preconditions:

- **User is logged in**

Steps:

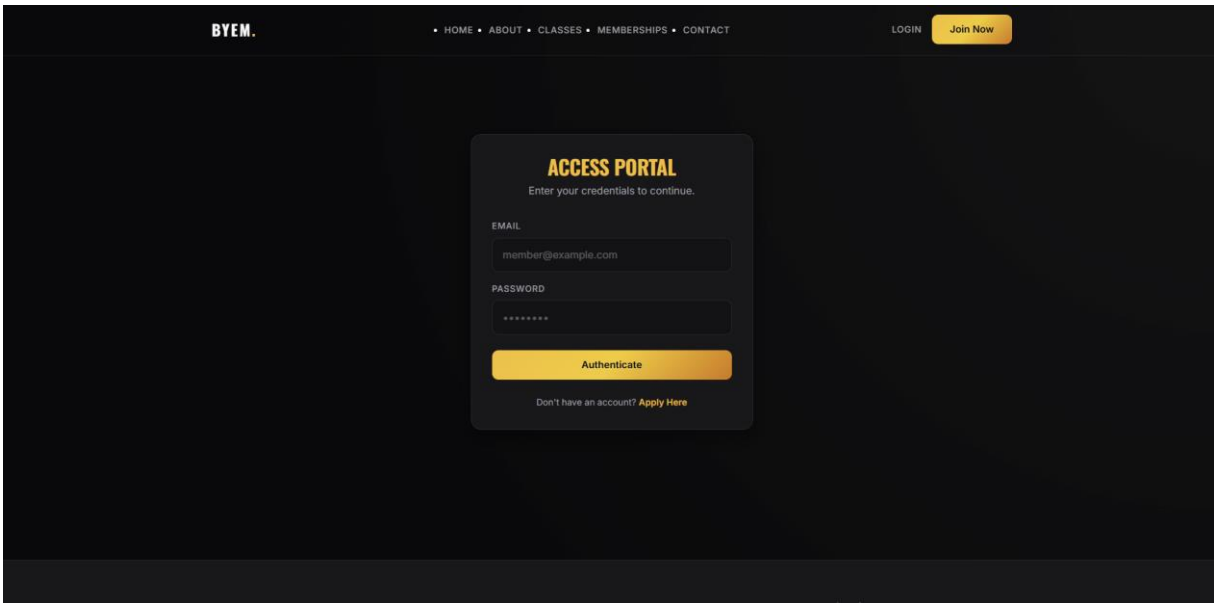
1. **Click Logout**
2. **Manually enter dashboard URL in browser**

Expected Result:
User should be logged out and direct dashboard access should be blocked.

Actual Result:
User cannot access dashboard after log out.

Status: PASS

Screenshot:



TC-09

Test Case ID: TC-09
Module: Booking
Title: Cancel Booking Successfully
Type: Functional Testing

Preconditions:

- **User is logged in**
- **User has an existing class reservation**
- **Dashboard or booking page open**

Steps:

1. **Open booked classes / dashboard page**
2. **Click Cancel Booking**
3. **Confirm cancellation if required**

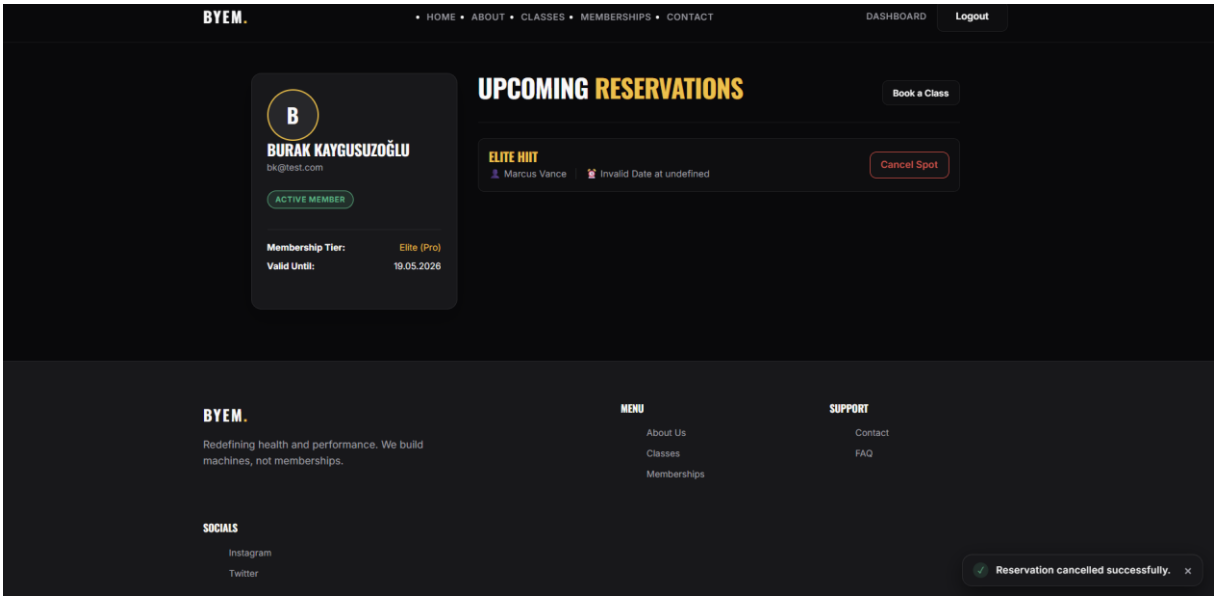
Test Data:
Previously reserved class

Expected Result:
Selected booking should be cancelled successfully and removed from reservation list.

Actual Result:
Reservation cancelled successfully.

Status: PASS

Screenshot:



BR-01

Bug ID: BR-01
Module: Footer UI
Title: Footer content appears incomplete and poorly balanced
Severity: Low
Priority: Medium

Preconditions:

- **Open Membership page**
- **Scroll to footer section**

Steps to Reproduce:

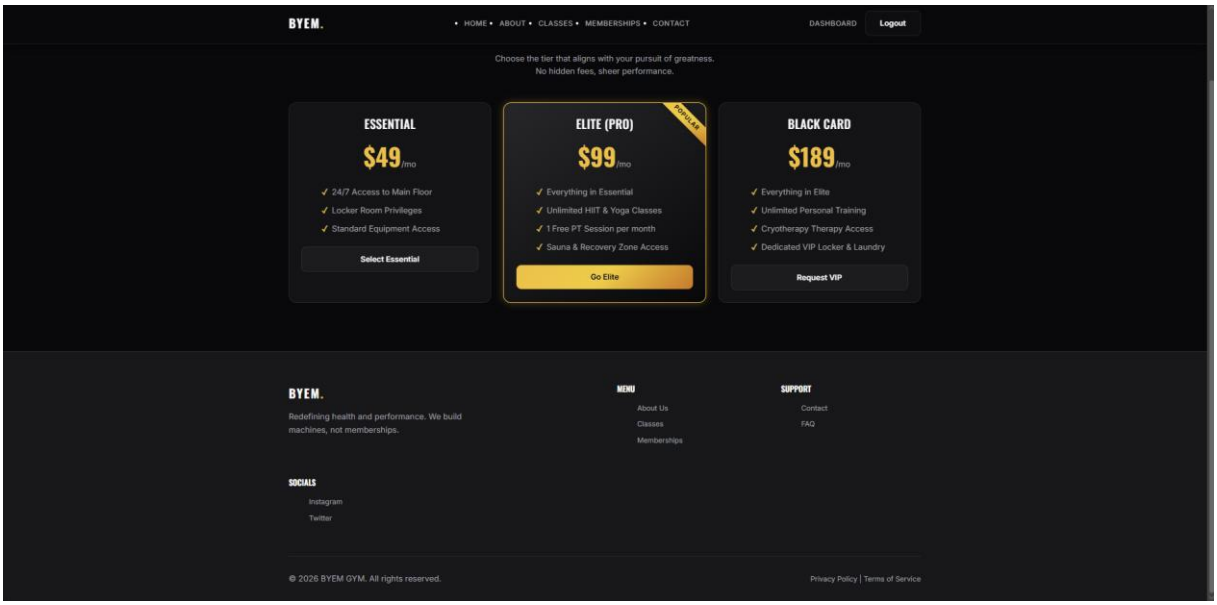
1. **Open Membership page**
2. **Scroll to bottom of page**
3. **Inspect footer layout and content spacing**

Expected Result:
Footer should contain balanced content, proper spacing, and complete section details.

Actual Result:
Footer contains limited content and appears visually incomplete with unbalanced spacing.

Status:
Open

Screenshot:



BR-02

Bug ID: BR-02
Module: Registration UI
Title: Registration form scales poorly under browser zoom levels
Severity: Low
Priority: Medium

Preconditions:

- **Open Register page**
- **Change browser zoom level**

Steps to Reproduce:

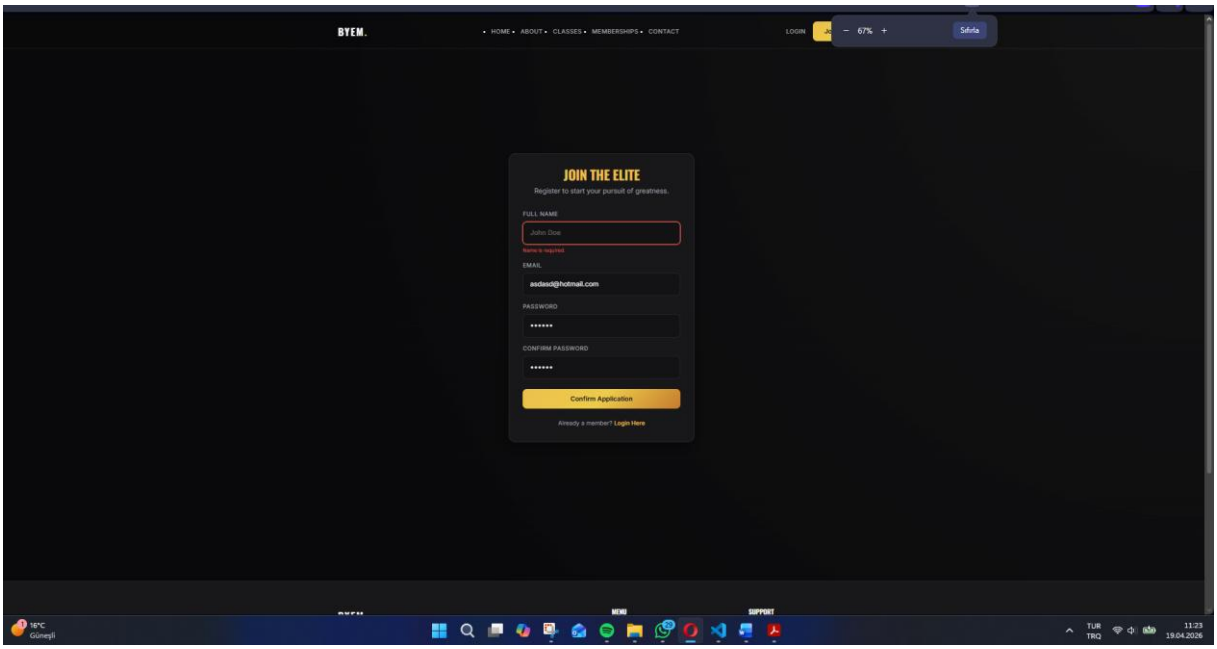
1. **Open Register page**
2. **Set browser zoom to 67%**
3. **Observe form size and layout spacing**

Expected Result:
Page layout should remain balanced and readable at common zoom levels.

Actual Result:
Form becomes too small and excessive empty space appears, reducing usability.

Status:
Open

Screenshot:



BR-03

Bug ID: BR-03
Module: Login UI
Title: Error message is not clear enough for failed login attempts
Severity: Low
Priority: Medium

Preconditions:

- **Login page open**

Steps to Reproduce:

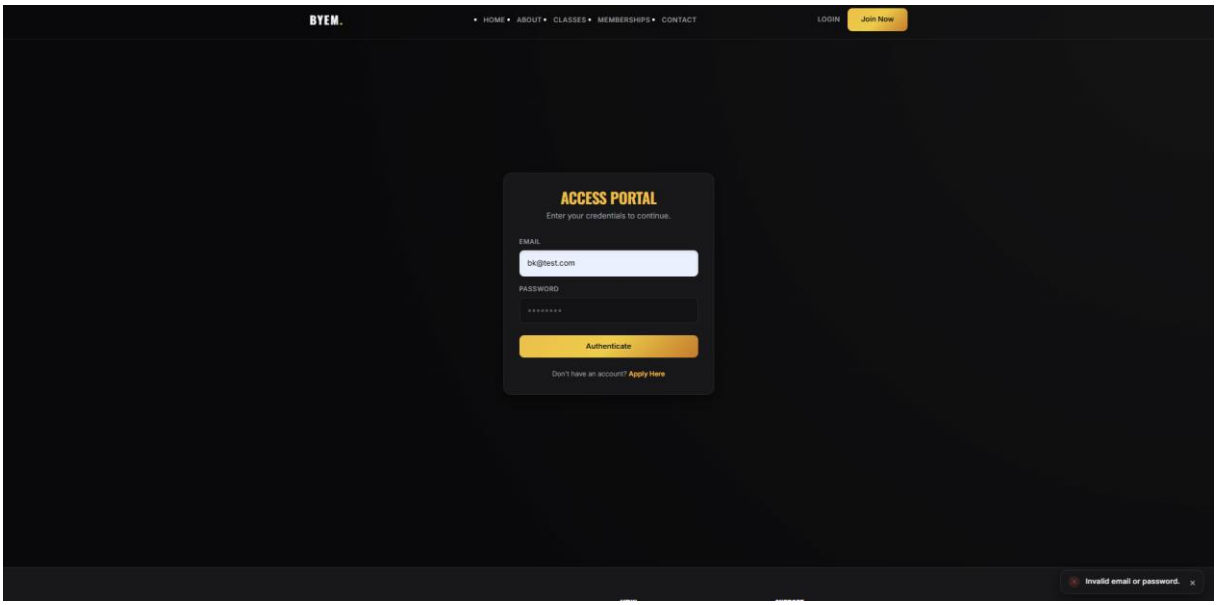
1. **Enter incorrect login credentials**
2. **Click Login**
3. **Observe displayed error message**

Expected Result:
System should show a clear, visible, and helpful error message.

Actual Result:
Displayed error message is unclear or provides insufficient guidance to the user.

Status:
Open

Screenshot:



BR-04

Bug ID: BR-04
Module: Classes / Booking UI
Title: Seat availability counter is unclear and may confuse users
Severity: Medium
Priority: Medium

Preconditions:

- **Open Classes page**

Steps to Reproduce:

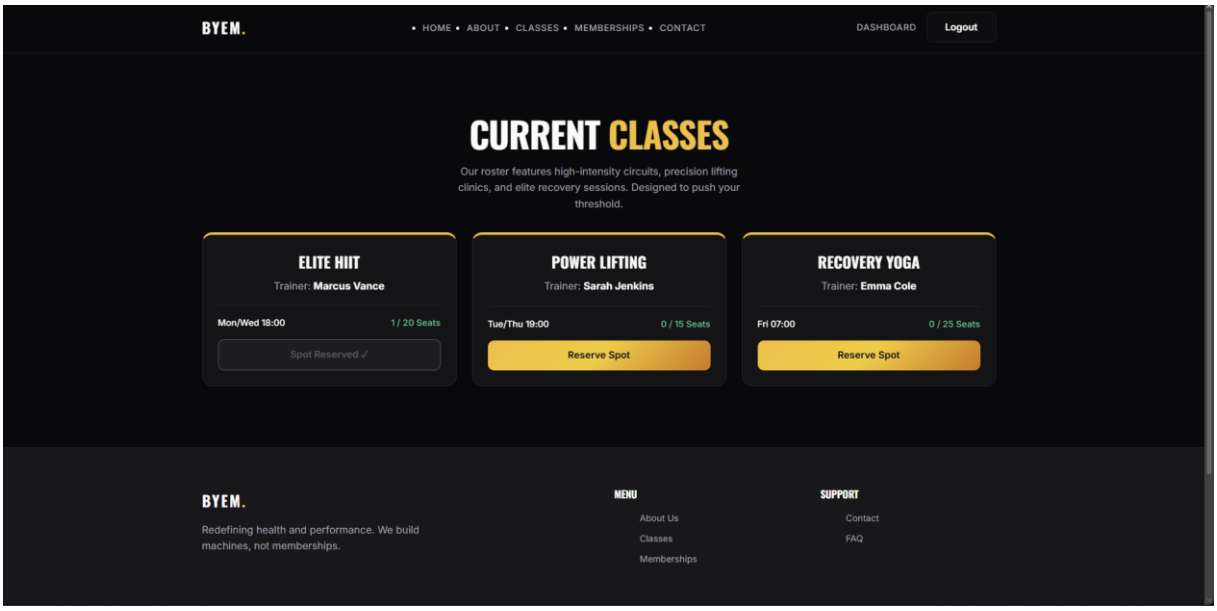
1. **Open Classes page**
2. **Observe seat information shown on class cards**
3. **Review values such as “0 / 15 Seats”**

Expected Result:
Seat information should clearly indicate booked seats, remaining seats, or total capacity.

Actual Result:
Displayed values are ambiguous and may confuse users about seat availability.

Status:
Open

Screenshot:



Advanced Bug Reports

Bug ID	Bug Title	Severity	Evidence Screenshot	Short Description
BR-05	LocalStorage Role Manipulation Risk	Critical	DevTools → Application → LocalStorage	User role/session data can be modified manually from browser storage.
BR-06	Payment Bypass Risk	High	Membership active screen + LocalStorage	Membership may be activated without secure server-side payment validation.
BR-07	Stored XSS Risk in User/Class Data	High	Input payload / rendered output / alert	Unsanitized user input may execute scripts when rendered in the page.
BR-08	Booking Capacity Inconsistency	Medium / High	Two tabs booking same class	Simultaneous booking attempts may create incorrect seat counts or duplicate reservations.

BR-05 — LocalStorage Role Manipulation Risk

Problem

Authentication and role data are stored client-side. A user may edit browser LocalStorage and change privileges manually.

Test Case

1. **Open website**
2. **Open DevTools**
3. **Go to Application → LocalStorage**
4. **Modify role/session value**
5. **Refresh page**

Expected Result

Role data should be validated securely by backend.

Actual Risk

Unauthorized access or privilege escalation may occur.

BR-06 — Payment Bypass Risk

Problem

Membership activation depends on client-side payment simulation rather than real secure payment confirmation.

Test Case

1. **Open membership page**
2. **Complete payment flow**
3. **Inspect LocalStorage / membership state**
4. **Try changing values manually**

Expected Result

Membership should activate only after verified payment transaction.

Actual Risk

Users may gain premium access without valid payment.

BR-07 — Stored XSS Risk in User/Class Data

Problem

If user input is inserted into DOM without sanitization, malicious scripts may run in future sessions.

Test Case

1. **Enter payload into Full Name field:**
   <img src=x onerror=alert(1)>
2. **Register user**
3. **Open dashboard / user profile**

Expected Result

Input should be escaped and displayed as plain text.

Actual Risk

JavaScript execution, session theft, page manipulation.

BR-08 — Booking Capacity Inconsistency

Problem

LocalStorage-based booking updates may fail under simultaneous multi-tab requests.

Test Case

1. **Open two browser tabs**
2. **Login same account**
3. **Open Classes page in both tabs**
4. **Reserve same class quickly**

Expected Result

Seat count should remain accurate and synchronized.

Actual Risk

Duplicate reservations or incorrect remaining capacity display.

Automation Testing

To improve testing efficiency and reduce repetitive manual execution, **Selenium WebDriver** was used for basic automation testing of the BYEM GYM Web-Based Management System.

Selenium was selected because it is a widely used industry-standard tool for browser automation and functional testing of web applications.

The automated tests focused on critical user authentication and validation scenarios.

Executed Automated Test Scenarios

- **AT-01 – Valid Login Test**
  Verified that a registered user can successfully log in using valid credentials and access the system.
- **AT-02 – Invalid Login Test**
  Verified that the system rejects incorrect login credentials and displays an appropriate error message.
- **AT-03 – Duplicate Registration Test**
  Verified that the system prevents a new registration attempt using an email address that already exists in the system.

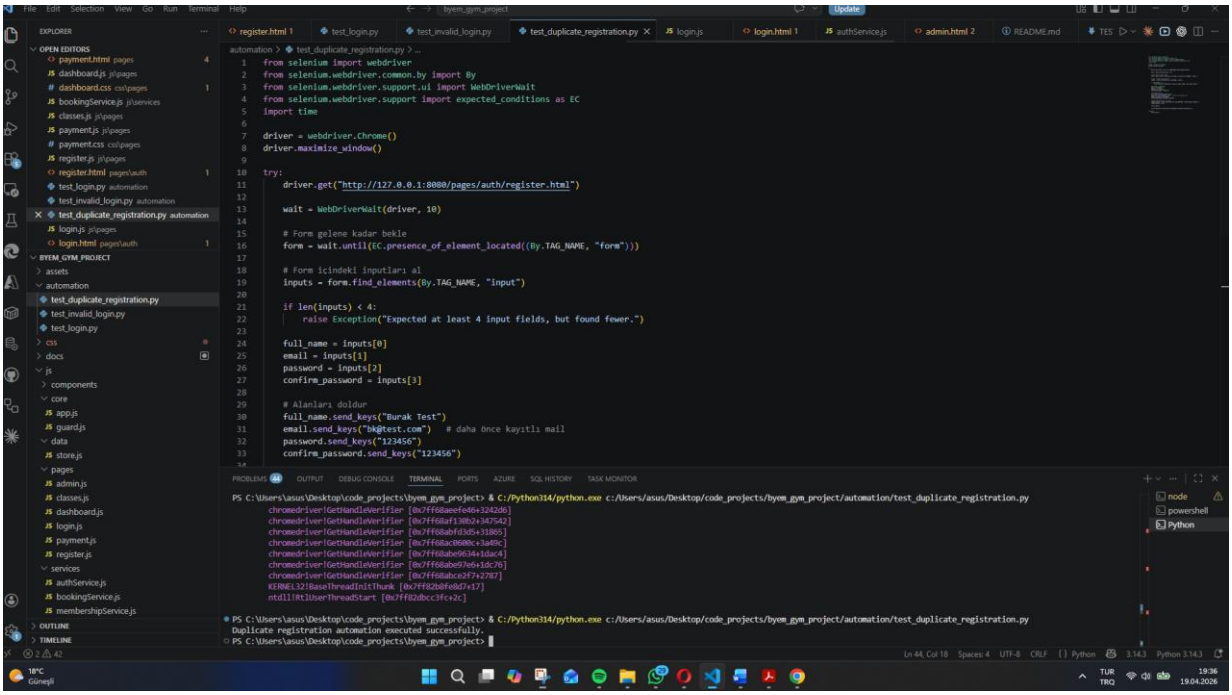
Test Execution Result

All Selenium automation scripts were executed successfully in the local test environment using Google Chrome browser.

Benefits of Automation Testing

- Saves time during repeated regression testing
- Reduces human error during test execution
- Provides faster verification of core system functions
- Improves overall software quality assurance process

AT-03: Duplicate Registration Testing



Performance Testing

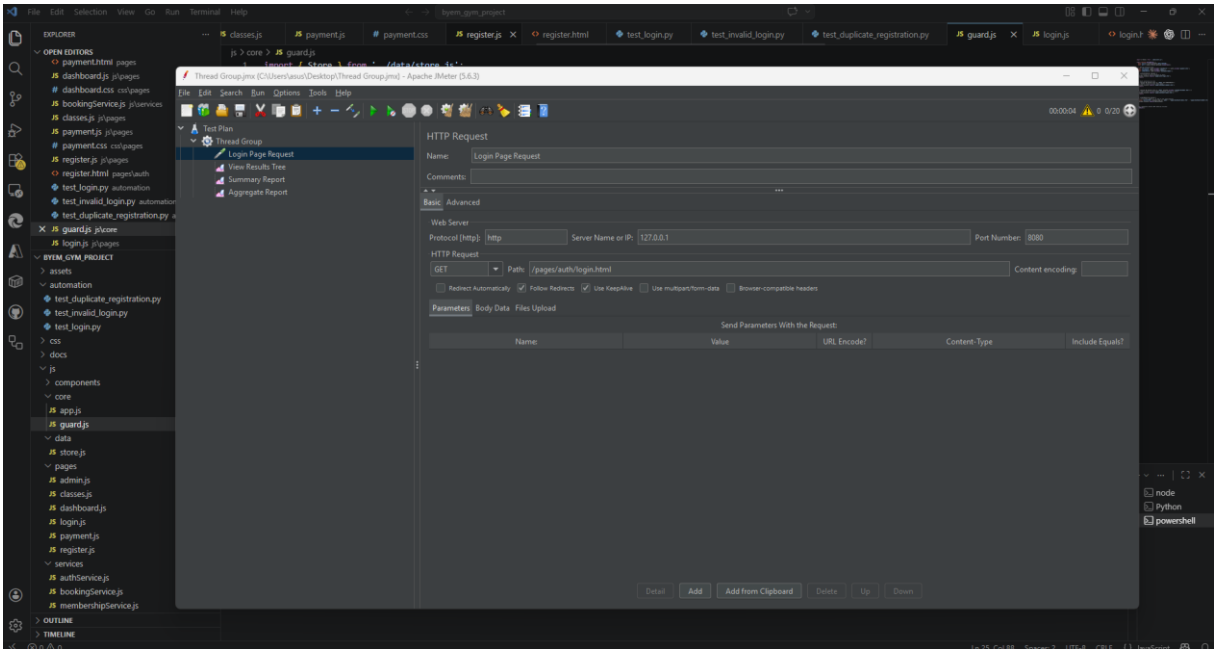
PT-01 LOGİN PAGE 30 USER THREADS 3 RAMPS

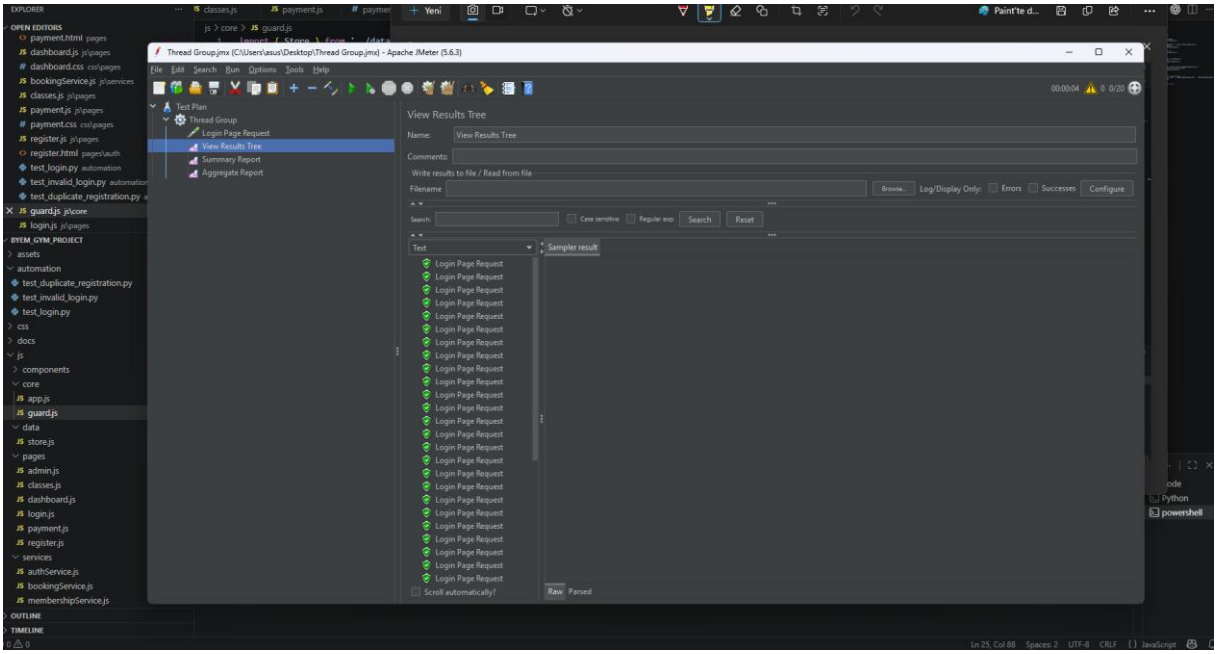
Performance Testing Results (Apache JMeter)

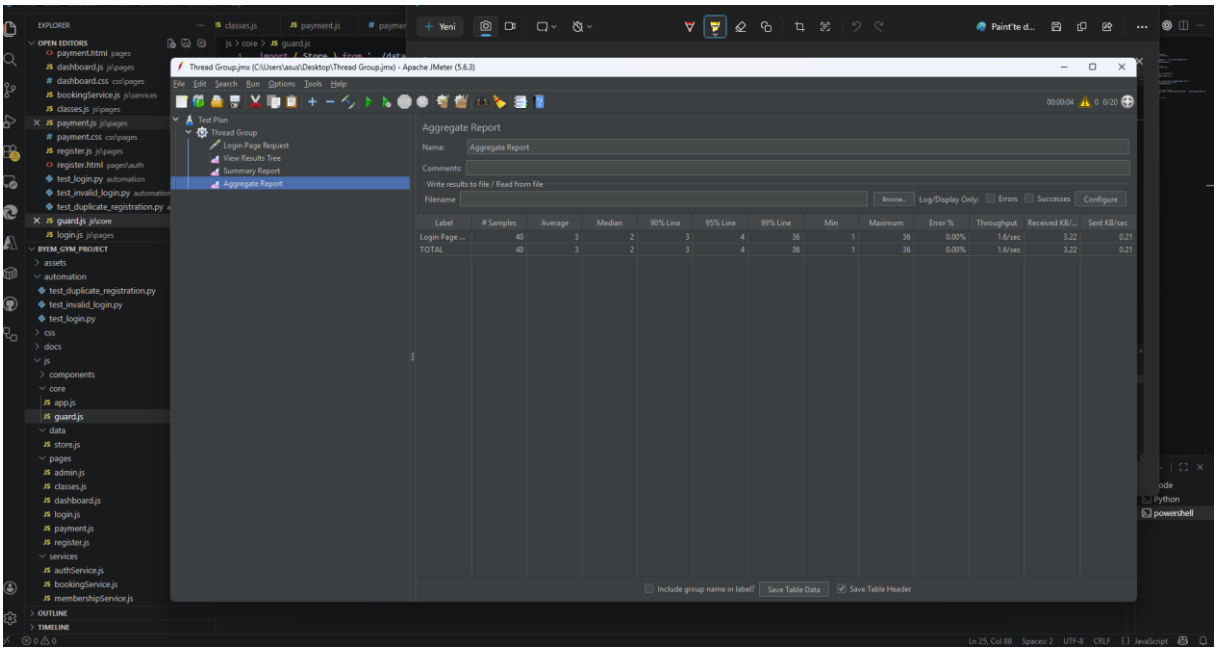
A concurrent load test was performed using Apache JMeter to evaluate system responsiveness under multiple user requests.

Parameter	Value
Scenario	Login Page Load Test
Virtual Users	30
Ramp-Up Time	5 seconds
Requests Executed	30
Average Response Time	3 ms
Maximum Response Time	45 ms
Error Rate	0.00%

Parameter	Value
Throughput	6.3 requests/sec







PT-02 LOGİN PAGE 100 NUMBER OF THREADS AND 5 RAMP

PT-02 — Spike Load Test

Purpose

To simulate a sudden traffic increase and observe system stability.

Important Summary Report Values

- **Users:** 100
- **Samples:** 100 requests completed
- **Average:** 2 ms
- **Max:** 3 ms
- **Error %:** 0.00%
- **Throughput:** 20.2 requests/sec

Result

The application handled sudden traffic successfully with no failures.

PT-03 — Higher Stress Test

Purpose

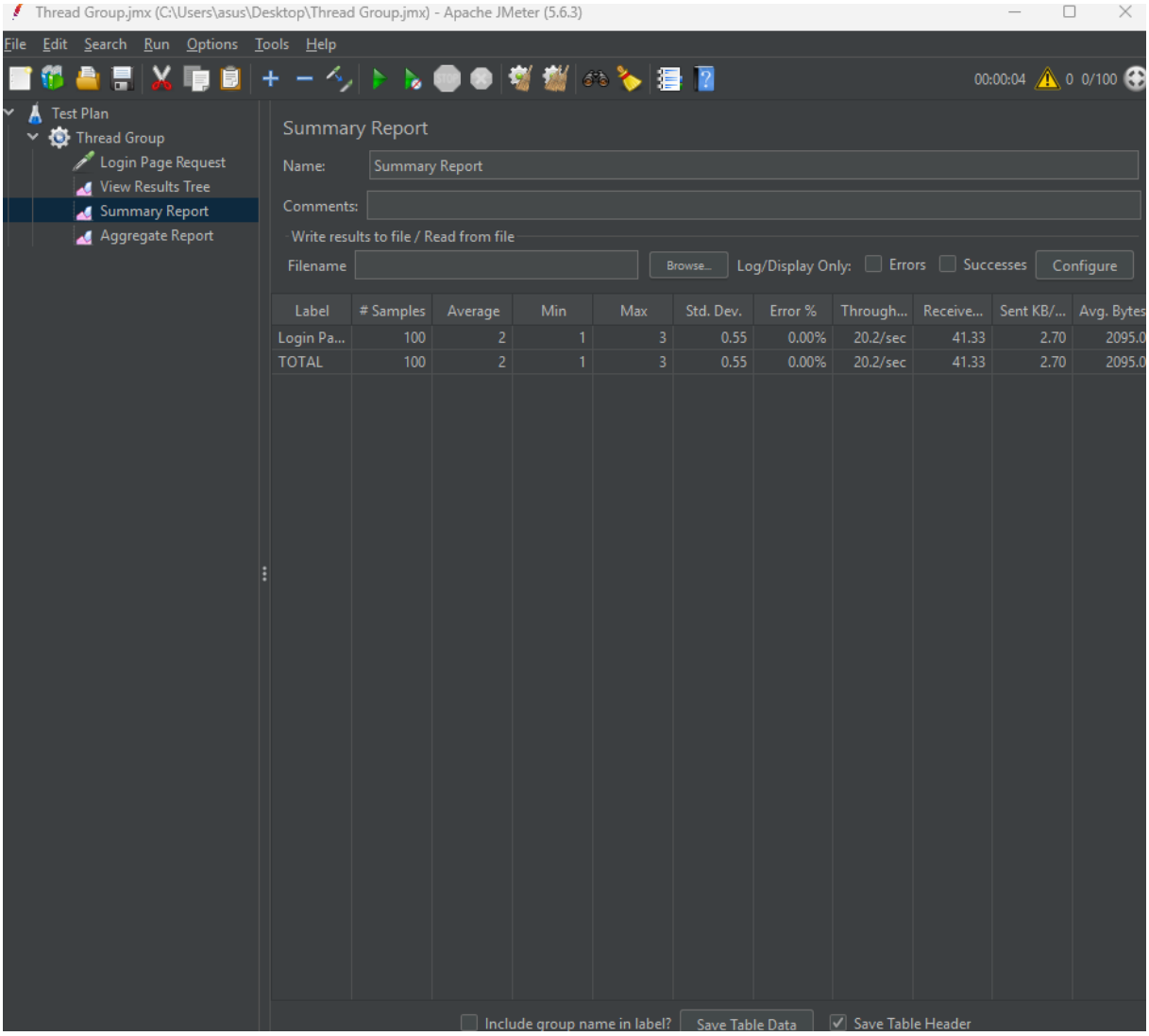
To test system behavior under a heavier user load.

Important Summary Report Values

- **Users:** 200
- **Samples:** 200 requests attempted
- **Average:** 2 ms
- **Max:** 4 ms
- **Error %:** 100.00%
- **Throughput:** 20.1 requests/sec

Result

All requests failed due to configuration/request issue during the test. This scenario highlighted the importance of correct request setup during stress testing.



PT-04 — Endurance Test

Purpose

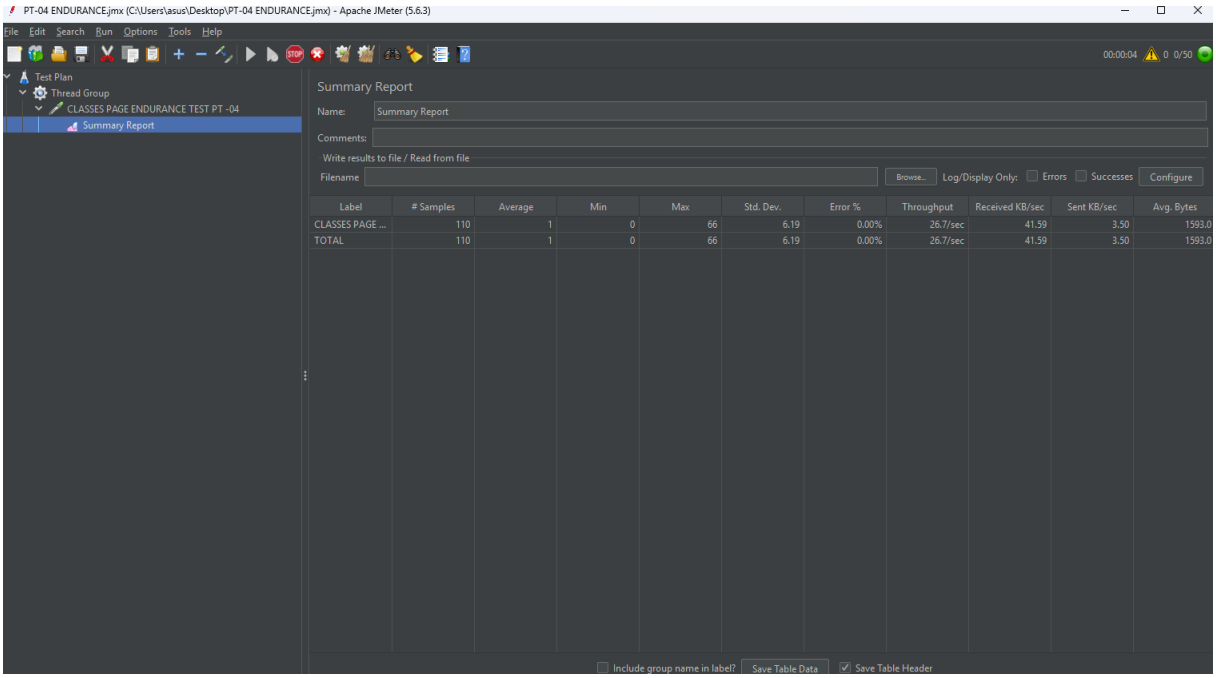
To test repeated request stability over time.

Important Summary Report Values

- **Users:** 50
- **Repeated Requests:** 110+ samples observed
- **Average:** 1 ms
- **Max:** 66 ms
- **Error %:** 0.00%
- **Throughput:** 26.7 requests/sec

Result

The application remained stable during repeated traffic simulation with no errors.



Overall Performance Conclusion

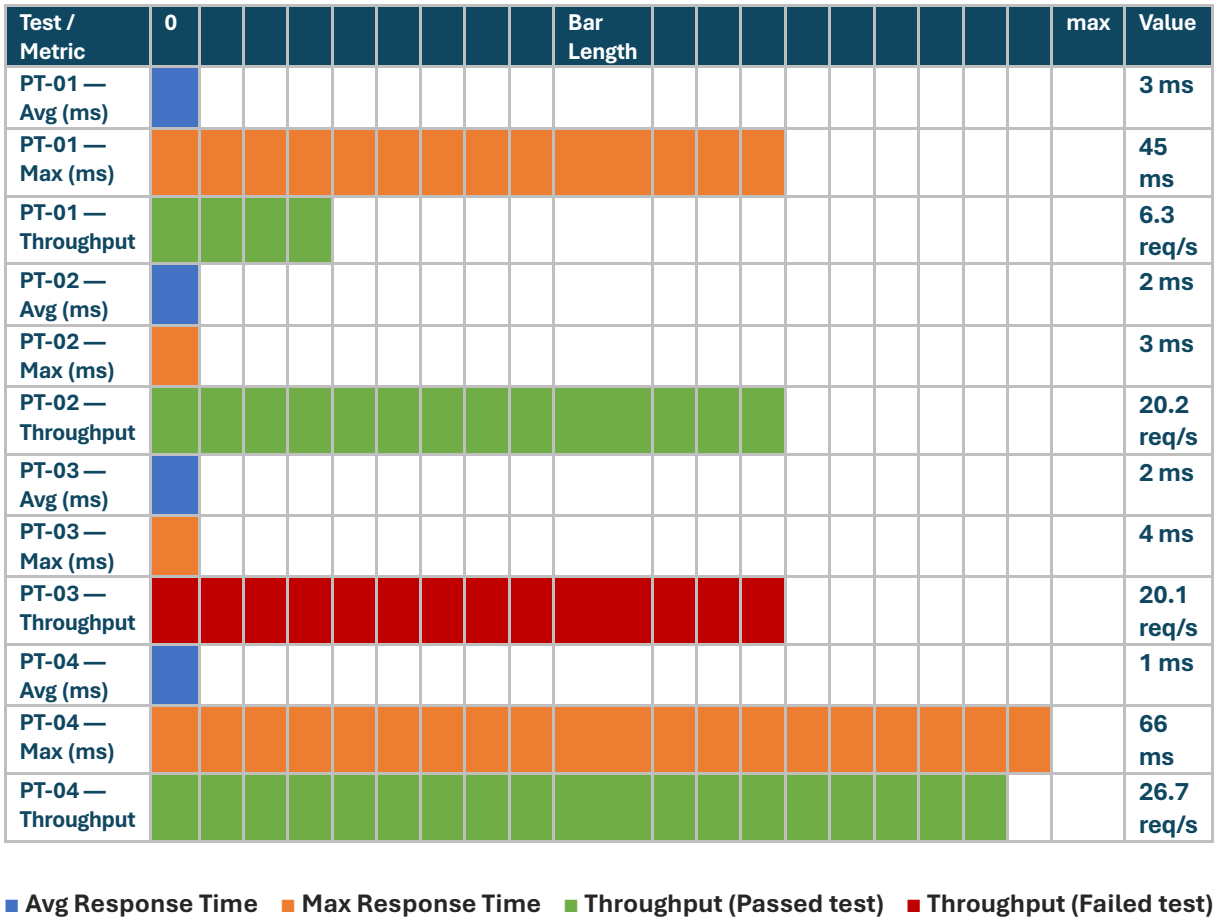
The BYEM GYM system showed strong performance in spike and endurance scenarios. Most successful tests completed with **0% error rate** and fast response times.

Performance Test Results — Comparative Summary

Test ID	Scenario	Users	Samples	Avg (ms)	Max (ms)	Error %	Throughput (req/s)	Status
PT-01	Login Page Load	30	30	3	45	0.00%	6.3	**Passed**
PT-02	Spike Load Test	100	100	2	3	0.00%	20.2	**Passed**
PT-03	Higher Stress Test	200	200	2	4	100.00%	20.1	**Failed**
PT-04	Endurance Test	50	110+	1	66	0.00%	26.7	**Passed**

Visual Performance Chart

Note: Avg/Max bars use a 0–70 ms scale; Throughput bars use a 0–30 req/sec scale. Bar lengths are comparable within the same metric, not across metrics.



ST-01

Test ID: ST-01
Test Title: Logout Session Validation (Back Button Access Test)
Test Type: Security Testing
Module: Authentication / Session Management
Severity: Medium
Status: Failed

Objective:
To verify whether the system fully terminates the user session after logout and prevents access to previously opened protected pages using the browser back button.

Preconditions:

- User account exists
- User is successfully logged in
- Dashboard page is accessible before logout

Test Steps:

1. Login with valid credentials
2. Open dashboard page
3. Click Logout button
4. After redirect, press browser Back button
5. Observe page behavior

Expected Result:
After logout, protected pages such as dashboard should not be accessible. System should redirect the user to login page.

Actual Result:
After pressing the browser back button, the dashboard page became visible again.

Conclusion:
This indicates that cached protected content remains accessible after logout. The application should fully clear session state and prevent cached page access after logout.

Security Impact:
A logged-out user may still view previously opened protected content through browser history.

Recommended Fix:

- Clear session data completely during logout
- Prevent caching on protected pages
- Revalidate authentication when page is reopened

- Test Case 08 can also be considered a basic security test, as it validates access restriction after user logout.

Conclusion

The BYEM GYM Web-Based Management System was successfully evaluated through functional, automation, performance, bug reporting, and security testing techniques.

The testing process confirmed that core system features such as registration, login, membership management, and booking operations function correctly under normal conditions.

In addition, advanced testing activities helped identify realistic security and usability risks, while performance tests demonstrated stable system behavior under concurrent user traffic.

Overall, the project provided valuable practical experience in applying real-world software quality assurance methods and improving system reliability.